

Mechatronics Tutorial Sheet 3: Discrete-time systems and the Z-transform

1. Consider the discrete-time causal LTI system

$$H(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}}$$

with region of convergence $|z| > \frac{1}{2}$.

- Find a difference equation relating the inputs $x[n]$ to the outputs $y[n]$ for this system.
- Find, using the inverse z-transform, the impulse and step responses for this system.
- Assume that $x[n]$ is an impulse. Using the initial- and final- value theorems, find the values $y[0]$ and $\lim_{n \rightarrow \infty} y[n]$. Verify that these are consistent with the impulse response calculated using the inverse z-transform.

$$(i) \quad H(z) = \frac{Y(z)}{X(z)}$$

$$\therefore (1 - \frac{1}{2}z^{-1})Y(z) = (z^{-1} - \frac{1}{2})X(z)$$

$$\therefore y[n] - \frac{1}{2}y[n-1] = x[n-1] - \frac{1}{2}x[n]$$

$$(ii) \quad Y(z) = H(z)X(z)$$

$$\text{for DT impulse, } X(z) = \sum_{n=0}^{\infty} \delta[n]z^{-n} = 1$$

$$\therefore Y(z) = H(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}} = \frac{2(\frac{1}{2}z^{-1} - 1) + \frac{3}{2}}{1 - \frac{1}{2}z^{-1}}$$

according to table, $|z| > \frac{1}{2}$

$$\begin{aligned} y[n] &= z^{-1} \left\{ -2 + \frac{\frac{3}{2}}{1 - \frac{1}{2}z^{-1}} \right\} \\ &= -2\delta[n] + \frac{3}{2}\left(\frac{1}{2}\right)^n u[n] \end{aligned}$$

	Signal, $x(n)$	z -Transform, $X(z)$	ROC
1	$\delta(n)$	1	All z
2	$u(n)$	$\frac{1}{1-z^{-1}}$	$ z > 1$
3	$a^n u(n)$	$\frac{1}{1-az^{-1}}$	$ z > a $
4	$na^n u(n)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z > a $
5	$-a^n u(-n-1)$	$\frac{1}{1-az^{-1}}$	$ z < a $
6	$-na^n u(-n-1)$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$ z < a $
7	$(\cos \omega_0 n) u(n)$	$\frac{1-z^{-1} \cos \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
8	$(\sin \omega_0 n) u(n)$	$\frac{z^{-1} \sin \omega_0}{1-2z^{-1} \cos \omega_0 + z^{-2}}$	$ z > 1$
9	$(a^n \cos \omega_0 n) u(n)$	$\frac{1-az^{-1} \cos \omega_0}{1-2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $
10	$(a^n \sin \omega_0 n) u(n)$	$\frac{az^{-1} \sin \omega_0}{1-2az^{-1} \cos \omega_0 + a^2 z^{-2}}$	$ z > a $

for DT unit step, $Y(z) = H(z) X(z)$

$$\therefore Y(z) = \frac{z^{-1} - \frac{1}{2}}{1 - \frac{1}{2}z^{-1}} \cdot \frac{1}{1-z^{-1}} = \frac{-\frac{3}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1-z^{-1}}$$

$$\therefore y[n] = \left(-\frac{3}{2}\left(\frac{1}{2}\right)^n + 1\right) u[n]$$

$$(iii) \quad y[n] = -2\delta[n] + \frac{3}{2}\left(\frac{1}{2}\right)^n u[n]$$

$$\text{for } y[0], \quad y[0] = \lim_{z \rightarrow \infty} Y(z) = \frac{-2 + \frac{3}{2}}{1} = -\frac{1}{2}$$

$$\text{for } \lim_{n \rightarrow \infty} y[n], \quad y[n] = \lim_{z \rightarrow 1} \{(z-1) Y(z)\} = 0$$

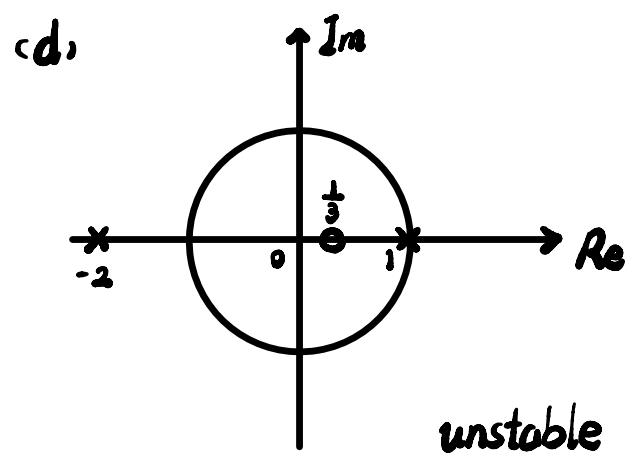
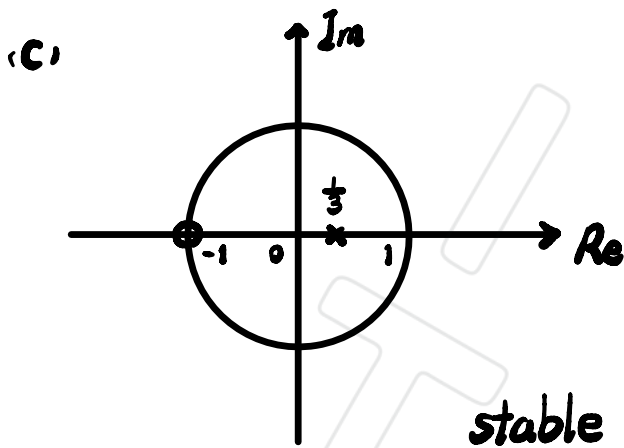
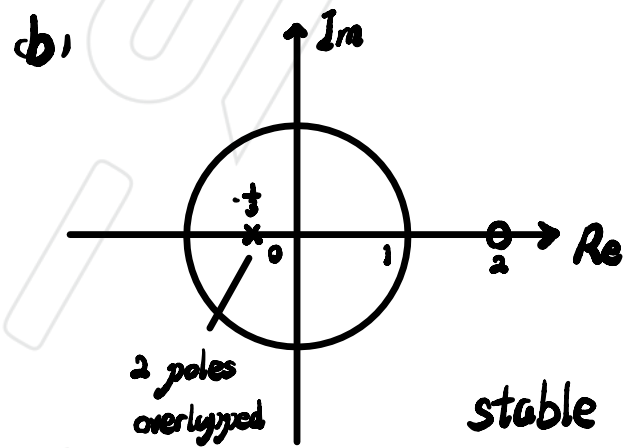
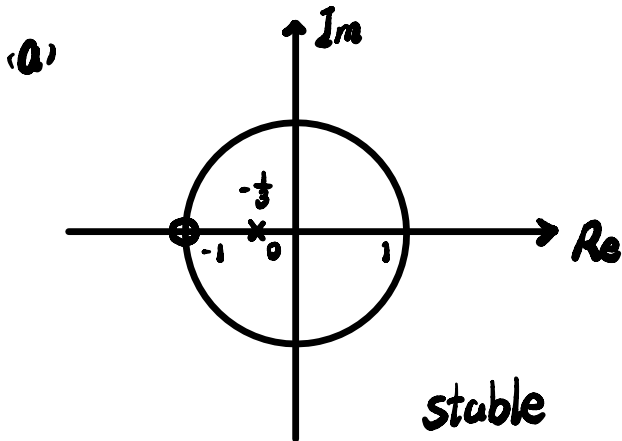
2. Draw the pole-zero diagram for each of the following discrete-time LTI systems and determine whether they are stable:

a) $H(z) = \frac{(1 + z^{-1})}{(1 + \frac{1}{3}z^{-1})}$

c) $H(z) = \frac{(\frac{1}{2} - z^{-1})}{(1 - \frac{1}{3}z^{-1})^2}$

b) $H(z) = \frac{(1 + z^{-1})}{(1 - \frac{1}{3}z^{-1})}$

d) $H(z) = \frac{(1 - \frac{1}{3}z^{-1})}{(1 - z^{-1})(1 + 2z^{-1})}$



3. A discrete-time system has transfer function

$$H(z) = z^{-1}.$$

- a) What function does the system serve? That is, what does the system do to its input?
- b) Is the system stable, unstable or marginally stable?
- c) The system is now interconnected in a feedback path with a second system with transfer function $K(z)$ as shown in the block diagram in Figure 1. What is the transfer function (from input $x[n]$ to output $y[n]$)?
- d) Suppose the controller has transfer function $K(z) = a$. What are the poles and zeros of the closed-loop system and for which values of a is the system stable, marginally stable and unstable?

a) A time delay of the input

b) stable (pole is $z = 0$)

c)

$$G(z) = \frac{H(z)}{1 - H(z)K(z)}$$

d)

$$G(z) = \frac{H(z)}{1 - aH(z)} = \frac{z^{-1}}{1 - az^{-1}} = \frac{1}{z - a}$$

zero : ~~$z = 0$~~ none

pole : $z = a$

$\therefore |a| < 1$, stable

$|a| = 1$, marginally stable

$|a| > 1$, unstable

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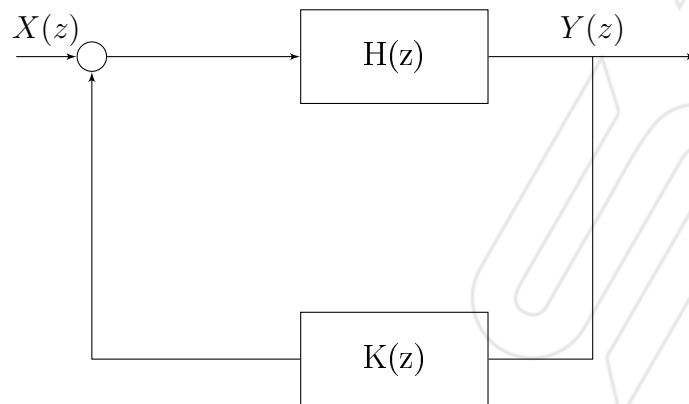


Figure 1: Block diagram representation of the closed-loop system in Question 3(c) and (d).

4. A system, let's call it system A, is described by the difference equation

$$y[n] - ay[n-1] = x[n].$$

A second system, let's call it system B, is described by the difference equation

$$y[n] = x[n] + ax[n].$$

The parameter a is some scalar quantity. Show that stability properties of system A depends on the parameter a and state for which values of a system A is stable, marginally stable and unstable. In contrast to this, show that system B is always stable.

for system A. $(1 - az^{-1})Y(z) = X(z)$

$$\therefore H(z) = \frac{Y(z)}{X(z)} = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}$$

$\therefore$ pole: $z = a$ $\therefore$ shown in previous question

for system B. $Y(z) = (1 + a)X(z)$

$\therefore H(z) = 1 + a$ $\therefore z$ can be any value $\therefore$ always stable